

Sheryl Paul

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Computer Science PhD Candidate

CPS-VIDA Lab

Thomas Lord Department of Computer Science

University of Southern California

Los Angeles, California

PROFESSIONAL SUMMARY

Computer Science PhD candidate studying planning, learning, and control of multi-agent and networked systems. Research combines adaptive control, formal methods, optimization, and evolutionary game theory to address uncertainty, coordination, and coupled constraints, with theoretical analysis and computational evaluation.

EDUCATION

PhD, Computer Science

University of Southern California

2019–December 2026 (expected)[†]

- Dissertation: *Planning and Control of Multi-Agent Systems*.

MSc, Computer Science

University of Oxford

2018–2019

2:1; 1:1

- Thesis: *The Emergence of Social Norms in Evolutionary Games*.

BE, Computer Engineering

University of Mumbai

2014–2018

GPA: 9.79/10.00

- Capstone: *Heterogeneous Clustering for Universal Recommendation Generation* (published patent).

RESEARCH EXPERIENCE

Doctoral Researcher, CPS-VIDA Lab

University of Southern California

2019–Present

Los Angeles, California

- Developed adaptive control and online parameter-estimation methods for opinion dynamics with unknown susceptibilities; established stability and convergence conditions and characterized performance under intervention budgets and finite horizons.
- Developed evolutionary game-theoretic methods for multi-agent planning with timed objectives and policy adaptation under environmental shifts.
- Developed MIP and SMT formulations for STL-GO planning under coupled spatial, temporal, and graph constraints, with soundness guarantees; ongoing work explores optimization and multi-agent reinforcement learning.
- Developed algebraic quantitative semantics and contributed to automata-based monitoring of spatio-temporal requirements over dynamic graphs.

Research Assistant, Department of Social Policy and Intervention

University of Oxford

2018–2019

Oxford, United Kingdom

- Scraped and analyzed electric-vehicle job-market and purchase data across the UK and EU to assess progress in the transition toward EV adoption.

TEACHING AND MENTORING

- Teaching Assistant, CSCI 513: Autonomous Cyber-Physical Systems:** USC (Spring 2025). Supported projects in modeling, control, formal specifications, verification, planning, and reinforcement learning using MATLAB/Simulink and Python.
- Research Mentor:** Mentored undergraduate and master's students on formal methods, robust spatio-temporal semantics, multi-agent planning, cyber-physical systems, and reinforcement learning.
- Teaching Assistant, CSCI 356: Computer Systems:** USC (Fall 2024). Taught systems programming in C, x86 assembly, memory hierarchy, virtual memory, compilation, and linking.
- Teaching Assistant, Structured Programming Approach:** University of Mumbai (2016–2017). Taught C programming.

TECHNICAL SKILLS

- Formal methods and verification:** STL, STREL, STL-GO, Z3, dReal, SMT solving, quantitative semantics, automata-based monitoring, and runtime verification.
- Robotics and multi-agent systems:** specification-guided planning, multi-agent coordination, partial observability, neuro-symbolic methods, and multi-agent reinforcement learning; Gymnasium, PettingZoo, SwarmLab, SUMO, Stable-Baselines3, Ray RLlib, and MARLLib.
- Optimization and control:** mixed-integer, convex, gradient-based, and derivative-free optimization; online control and parameter estimation; Gurobi, CVXPY, SciPy, and CMA-ES.
- Programming and scientific computing:** Python, MATLAB/Simulink, C/C++, NumPy, pandas, scikit-learn, PyTorch, and TensorFlow/Keras; HPC-cluster workflows and Git.

[†] Includes a voluntary leave of absence.

PEER-REVIEWED PUBLICATIONS

First or joint-first author on 8 of 13 listed publications and manuscripts; ** denotes equal contribution.

1. **On Online Control of Opinion Dynamics.** S. Paul, L. C. Juarez, J. V. Deshmukh, and K. Savla. *American Control Conference (ACC)*, 2026.
2. **Multi-agent Path Finding for Timed Tasks using Evolutionary Games.** S. Paul, A. Balakrishnan, X. Qin, and J. V. Deshmukh. *QEST-FORMATS*, 2024.
3. **Multi-Agent Planning with Spatio-Temporal and Topological Constraints using STL-GO.** S. Paul**, V. Kudalkar**, A. Balakrishnan, L. Lindemann, A. Speranzon, and J. V. Deshmukh. *Formal Methods in Computer-Aided Design (FMCAD)*, 2026.
4. **Survival of the Fittest: Evolutionary Adaptation of Policies for Environmental Shifts.** S. Paul and J. V. Deshmukh. *European Conference on Artificial Intelligence (ECAI)*, 2024.
5. **Monitoring Spatially Distributed Cyber-Physical Systems with Alternating Finite Automata.** A. Balakrishnan, S. Paul, S. Silveti, L. Nenzi, and J. V. Deshmukh. *Hybrid Systems: Computation and Control (HSCC)*, 2025. *Best Paper Award; extended version under submission to NAHS.*
6. **Agent-Based Evolutionary Dynamics for Mixed Autonomy Weaving Ramps.** S. Paul, K. Wang, R. Li, and J. V. Deshmukh. *CPHS*, 2026.
7. **An Algebraic Framework for Quantitative Semantics of Spatio-Temporal Logic with Graph Operators.** S. Paul**, V. Kudalkar**, A. Balakrishnan**, T. Wu, L. Lindemann, and J. V. Deshmukh. *QEST-FORMATS*, 2026.
8. **Neuro-Symbolic Planning for Multi-Agent Systems using Differentiable Spatio-Temporal Logic.** V. Kudalkar, S. Paul, N. Hashemi, and J. V. Deshmukh. *NeuS*, 2026.
9. **Systematic Translation from Natural Language Robot Task Descriptions to STL.** S. Mohammadinejad, S. Paul, Y. Xia, V. Kudalkar, J. Thomason, and J. V. Deshmukh. *AISoLA*, 2024.
10. **Detection of Breast Cancer from Mammograms using a Hybrid Approach of Deep Learning and Linear Classification.** S. Agrawal, R. Rangnekar, D. Gala, S. Paul, and D. Kalbande. *IEEE International Conference on System Engineering and Technology (ICSET)*, 2018.

MANUSCRIPTS UNDER REVIEW

11. **Multi-agent Planning in Stochastic Dynamic Environments under Partial Observability Guided by Spatio-temporal Logic Requirements.** V. Kudalkar**, S. Paul**, S. Bharambe, S. Shim, M. Castillo-Effen, A. Speranzon, S. Varadarajan, and J. V. Deshmukh. *Under submission.*
12. **Monitoring Spatially Distributed Cyber-Physical Systems with Alternating Finite Automata (Extended Journal Version).** A. Balakrishnan, S. Paul, S. Silveti, L. Nenzi, and J. V. Deshmukh. *Under submission to Nonlinear Analysis: Hybrid Systems (NAHS)*, 2026.
13. **Topology-Aware Congestion Pricing: Demand-Robust Routing using the Forman-Ricci Curvature.** S. Paul, S. J. Williams, K. Wang, X. Qin, R. Li, and J. V. Deshmukh. *Under submission, 2026.*

INDUSTRY EXPERIENCE

Technical Analyst Intern, Credit Suisse Investment Bank (2017).

Technical Intern, Gromor Fintech (2016–2017).

PRESENTATIONS AND ACADEMIC SERVICE

- **Poster presentations:** GC Women in CS Conference (2023); USC-AAI (2023, 2024, 2025).
- **Journal reviewer:** IEEE Transactions on Automation Science and Engineering.
- **Conference reviewer:** RV, L4DC, VMCAI, AAAI, ICCPS, CAV, ACC, and CDC.
- **Conference service:** volunteer, NASA Formal Methods (2022, 2026) and L4DC/NeuS (2026).

ADDITIONAL VOLUNTEER WORK

TAPkids Teacher (2022–26); Union Rescue Mission; School on Wheels; Dream Center LA; Rain of Mercy; Mentor for Change; Nanoom Rehab Center.

HONORS AND AWARDS

- **Best Paper Award**, HSCC 2025.
- **British Council Scholar**, full-tuition MSc award, University of Oxford, 2018–2019.